

$$\Rightarrow x = \frac{2}{3} - \frac{4}{3}i \text{ or } x = \frac{2}{3} + \frac{4}{3}i$$

Hence, the roots of the given equation are $\frac{2}{3} - \frac{4}{3}i$ and $\frac{2}{3} + \frac{4}{3}i$.

EXAMPLE 4 Solve the quadratic equation $2x^2 - 4x + 3 = 0$ by using the general expressions for the roots of a quadratic equation.

SOLUTION Comparing the given equation with the general form $ax^2 + bx + c = 0$, we get
 $a = 2, b = -4$ and $c = 3$

Substituting the values of a, b, c in $\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$, we get

$$\alpha = \frac{4 + \sqrt{16 - 24}}{4} \quad \text{and,} \quad \beta = \frac{4 - \sqrt{16 - 24}}{4}$$

$$\Rightarrow \alpha = \frac{4 + \sqrt{-8}}{4} \quad \text{and,} \quad \beta = \frac{4 - \sqrt{-8}}{4}$$

$$\Rightarrow \alpha = \frac{4 + 2\sqrt{2}i}{4} \quad \text{and,} \quad \beta = \frac{4 - 2\sqrt{2}i}{4}$$

$$\Rightarrow \alpha = 1 + \frac{1}{\sqrt{2}}i \quad \text{and,} \quad \beta = 1 - \frac{1}{\sqrt{2}}i$$

Hence, the roots of the given equation are $1 + \frac{1}{\sqrt{2}}i$ and $1 - \frac{1}{\sqrt{2}}i$.

EXAMPLE 5 Solve the equation $25x^2 - 30x + 11 = 0$ by using the general expression for the roots of a quadratic equation.

SOLUTION Comparing the given equation with the general form of the quadratic equation $ax^2 + bx + c = 0$, we get: $a = 25, b = -30$ and $c = 11$.

Substituting these values in $\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$, we get

$$\alpha = \frac{30 + \sqrt{900 - 1100}}{50} \quad \text{and} \quad \beta = \frac{30 - \sqrt{900 - 1100}}{50}$$

$$\Rightarrow \alpha = \frac{30 + \sqrt{-200}}{50} \quad \text{and} \quad \beta = \frac{30 - \sqrt{-200}}{50}$$

$$\Rightarrow \alpha = \frac{30 + 10i\sqrt{2}}{50} \quad \text{and} \quad \beta = \frac{30 - 10i\sqrt{2}}{50}$$

$$\Rightarrow \alpha = \frac{3}{5} + \frac{\sqrt{2}}{5}i \quad \text{and} \quad \beta = \frac{3}{5} - \frac{\sqrt{2}}{5}i$$

Hence, the roots of the given equation are $\frac{3}{5} \pm \frac{\sqrt{2}}{5}i$.

EXERCISE 14.1

LEVEL-1

Solve the following quadratic equations by factorization method only (1-5):

1. $x^2 + 1 = 0$

2. $9x^2 + 4 = 0$

3. $x^2 + 2x + 5 = 0$

4. $4x^2 - 12x + 25 = 0$

5. $x^2 + x + 1 = 0$

Solve the following quadratics (6-18):

6. $4x^2 + 1 = 0$

8. $x^2 + 2x + 2 = 0$

10. $21x^2 + 9x + 1 = 0$

12. $x^2 + x + 1 = 0$ [NCERT]

14. $27x^2 - 10x + 1 = 0$ [NCERT]

16. $21x^2 - 28x + 10 = 0$ [NCERT]

18. $13x^2 + 7x + 1 = 0$

20. $\sqrt{3}x^2 - \sqrt{2}x + 3\sqrt{3} = 0$ [NCERT]

22. $x^2 + x + \frac{1}{\sqrt{2}} = 0$ [NCERT]

24. $\sqrt{5}x^2 + x + \sqrt{5} = 0$ [NCERT]

26. $x^2 - 2x + \frac{3}{2} = 0$ [NCERT]

7. $x^2 - 4x + 7 = 0$

9. $5x^2 - 6x + 2 = 0$

11. $x^2 - x + 1 = 0$

13. $17x^2 - 8x + 1 = 0$

15. $17x^2 + 28x + 12 = 0$

17. $8x^2 - 9x + 3 = 0$

19. $2x^2 + x + 1 = 0$

21. $\sqrt{2}x^2 + x + \sqrt{2} = 0$ [NCERT]

23. $x^2 + \frac{x}{\sqrt{2}} + 1 = 0$ [NCERT]

25. $-x^2 + x - 2 = 0$ [NCERT]

27. $3x^2 - 4x + \frac{20}{3} = 0$ [NCERT]

ANSWERS

1. $i, -i$

2. $\frac{2}{3}i, -\frac{2}{3}i$

3. $-1 + 2i, -1 - 2i$

4. $\frac{3}{2} + 2i, \frac{3}{2} - 2i$

5. $-\frac{1}{2} + i\frac{\sqrt{3}}{2}, -\frac{1}{2} - i\frac{\sqrt{3}}{2}$

6. $\frac{1}{2}i, -\frac{1}{2}i$

7. $2 \pm \sqrt{3}i$

8. $-1 \pm i$

9. $\frac{3}{5} \pm \frac{1}{5}i$

10. $\frac{-3}{14} \pm \frac{i\sqrt{3}}{42}$

11. $\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$

12. $-\frac{1}{2} \pm \frac{i\sqrt{3}}{2}$

13. $\frac{4}{17} \pm \frac{1}{17}i$

14. $\frac{5}{27} \pm \frac{\sqrt{2}}{27}i$

15. $\frac{-14}{17} \pm \frac{2\sqrt{2}}{17}i$

16. $\frac{2}{3} \pm \frac{\sqrt{14}}{21}i$

17. $\frac{9}{16} \pm \frac{\sqrt{15}}{16}i$

18. $\frac{-7}{26} \pm \frac{\sqrt{3}}{26}i$

19. $\frac{-1 \pm \sqrt{7}i}{4}$

20. $\frac{\sqrt{2} \pm \sqrt{34}i}{2\sqrt{3}}$

21. $\frac{-1 \pm \sqrt{7}i}{2\sqrt{2}}$

22. $\frac{-1 \pm \sqrt{2\sqrt{2} - 1}i}{2}$

23. $\frac{-1 \pm \sqrt{7}i}{2\sqrt{2}}$

24. $\frac{-1 \pm \sqrt{19}i}{2\sqrt{5}}$

25. $\frac{-1 \pm \sqrt{7}i}{-2}$

26. $1 \pm \frac{1}{\sqrt{2}}i$

27. $\frac{2}{3} \pm \frac{4}{3}i$

HINTS TO NCERT & SELECTED PROBLEMS

5. We have,

$$x^2 + x + 1 = 0$$

$$\Rightarrow x^2 + 2(x)\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + 1 = 0$$

$$\Rightarrow \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} = 0$$

$$\Rightarrow \left(x + \frac{1}{2}\right)^2 - \frac{3}{4}i^2 = 0$$

$$\Rightarrow \left(x + \frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}i\right)^2 = 0$$

$$\Rightarrow \left(x + \frac{1}{2} + \frac{\sqrt{3}}{2}i\right)\left(x + \frac{1}{2} - \frac{\sqrt{3}}{2}i\right) = 0$$

$$\Rightarrow \left(x + \frac{1 + \sqrt{3}i}{2}\right)\left(x + \frac{1 - \sqrt{3}i}{2}\right) = 0$$

$$\Rightarrow x + \frac{1 + \sqrt{3}i}{2} = 0 \text{ or, } x + \frac{1 - \sqrt{3}i}{2} = 0$$

$$\Rightarrow x = -\frac{1 + i\sqrt{3}}{2} \text{ or, } x = -\frac{1 - i\sqrt{3}}{2}$$

12. We have, $x^2 + x + 1 = 0$

Comparing the given equation with the general form $ax^2 + bx + c = 0$, we get

$$a = 1, b = 1 \text{ and } c = 1$$

Substituting the values of a, b, c in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{-1 + \sqrt{1 - 4}}{2} \text{ and } \beta = \frac{-1 - \sqrt{1 - 4}}{2}$$

$$\Rightarrow \alpha = \frac{-1 + i\sqrt{3}}{2} \text{ and } \beta = \frac{-1 - i\sqrt{3}}{2}$$

14. We have, $27x^2 - 10x + 1 = 0$

Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = 27, b = -10, c = 1$

Substituting the values of a, b, c in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\Rightarrow \alpha = \frac{10 + \sqrt{100 - 108}}{54} \text{ and } \beta = \frac{10 - \sqrt{100 - 108}}{54}$$

$$\Rightarrow \alpha = \frac{10 + \sqrt{-8}}{54} \text{ and } \beta = \frac{10 - \sqrt{-8}}{54}$$

$$\Rightarrow \alpha = \frac{5 + i\sqrt{2}}{27} \text{ and } \beta = \frac{5 - i\sqrt{2}}{27}$$

16. The given equation is $21x^2 - 28x + 10 = 0$. Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = 21, b = -28, c = 10$

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{28 + \sqrt{784 - 840}}{42} \text{ and } \beta = \frac{28 - \sqrt{784 - 840}}{42}$$

$$\Rightarrow \alpha = \frac{28 + \sqrt{-56}}{42} \text{ and } \beta = \frac{28 - \sqrt{-56}}{42}$$

$$\Rightarrow \alpha = \frac{2}{3} + \frac{i\sqrt{14}}{21} \text{ and } \beta = \frac{2}{3} - \frac{i\sqrt{14}}{21}$$

20. The given equation is $\sqrt{3}x^2 - \sqrt{2}x + 3\sqrt{3} = 0$. Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = \sqrt{3}$, $b = -\sqrt{2}$, $c = 3\sqrt{3}$.

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{\sqrt{2} + \sqrt{2 - 36}}{2\sqrt{3}} \text{ and } \beta = \frac{\sqrt{2} - \sqrt{2 - 36}}{2\sqrt{3}}$$

$$\Rightarrow \alpha = \frac{\sqrt{2} + i\sqrt{34}}{2\sqrt{3}} \text{ and } \beta = \frac{\sqrt{2} - i\sqrt{34}}{2\sqrt{3}}$$

21. The given equation is $\sqrt{2}x^2 + x + \sqrt{2} = 0$. Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = \sqrt{2}$, $b = 1$, $c = \sqrt{2}$.

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{-1 + \sqrt{1 - 8}}{2\sqrt{2}} \text{ and } \beta = \frac{-1 - \sqrt{1 - 8}}{2\sqrt{2}}$$

$$\Rightarrow \alpha = \frac{-1 + i\sqrt{7}}{2\sqrt{2}} \text{ and } \beta = \frac{-1 - i\sqrt{7}}{2\sqrt{2}}$$

22. The given equation is $x^2 + x + \frac{1}{\sqrt{2}} = 0$. Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = 1$, $b = 1$, $c = \frac{1}{\sqrt{2}}$.

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\Rightarrow \alpha = \frac{-1 + \sqrt{1 - 2\sqrt{2}}}{2} \text{ and } \beta = \frac{-1 - \sqrt{1 - 2\sqrt{2}}}{2}$$

$$\Rightarrow \alpha = \frac{-1 + i\sqrt{2\sqrt{2} - 1}}{2} \text{ and } \beta = \frac{-1 - i\sqrt{2\sqrt{2} - 1}}{2}$$

23. The given equation is $x^2 + \frac{x}{\sqrt{2}} + 1 = 0$. Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = 1$, $b = \frac{1}{\sqrt{2}}$ and $c = 1$.

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{-\frac{1}{\sqrt{2}} + \sqrt{\frac{1}{2} - 4}}{2} \text{ and } \beta = \frac{-\frac{1}{\sqrt{2}} - \sqrt{\frac{1}{2} - 4}}{2}$$

$$\Rightarrow \alpha = \frac{-1 + i\sqrt{7}}{2\sqrt{2}} \text{ and } \beta = \frac{-1 - i\sqrt{7}}{2\sqrt{2}}$$

24. The given equation is $\sqrt{5}x^2 + x + \sqrt{5} = 0$. Comparing this equation with $ax^2 + bx + c$, we get $a = \sqrt{5}$, $b = 1$ and $c = \sqrt{5}$. Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{-1 + \sqrt{1 - 20}}{2\sqrt{5}} \text{ and } \beta = \frac{-1 - \sqrt{1 - 20}}{2\sqrt{5}}$$

$$\Rightarrow \alpha = \frac{-1 + i\sqrt{19}}{2\sqrt{5}} \text{ and } \beta = \frac{-1 - i\sqrt{19}}{2\sqrt{5}}$$

25. The given equation is $-x^2 + x - 2 = 0$. Comparing this equation with $ax^2 + bx + c$, we get: $a = -1$, $b = 1$ and $c = -2$. Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\Rightarrow \alpha = \frac{-1 + \sqrt{1 - 8}}{-2} \text{ and } \beta = \frac{-1 - \sqrt{1 - 8}}{-2}$$

$$\Rightarrow \alpha = \frac{-1 + i\sqrt{7}}{-2} \text{ and } \beta = \frac{-1 - i\sqrt{7}}{-2}$$

$$\Rightarrow \alpha = \frac{1}{2} - i\frac{\sqrt{7}}{2} \text{ and } \beta = \frac{1}{2} + i\frac{\sqrt{7}}{2}$$

26. The given equation is $x^2 - 2x + \frac{3}{2} = 0$. Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = 1$, $b = -2$ and $c = 3/2$.

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow \alpha = \frac{2 + \sqrt{4 - 6}}{2} \text{ and } \beta = \frac{2 - \sqrt{4 - 6}}{2}$$

$$\Rightarrow \alpha = 1 + \frac{i}{\sqrt{2}} \text{ and } \beta = 1 - \frac{i}{\sqrt{2}}$$

27. The given equation is $3x^2 - 4x + \frac{20}{3} = 0$. Comparing this equation with $ax^2 + bx + c = 0$, we get: $a = 3$, $b = -4$ and $c = \frac{20}{3}$. Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{4 + \sqrt{16 - 80}}{6} \text{ and } \beta = \frac{4 - \sqrt{16 - 80}}{6}$$

$$\Rightarrow \alpha = \frac{4+8i}{6} \text{ and } \beta = \frac{4-8i}{6}$$

$$\Rightarrow \alpha = \frac{2}{3} + \frac{4}{3}i \text{ and } \beta = \frac{2}{3} - \frac{4}{3}i$$

14.5 QUADRATIC EQUATIONS WITH COMPLEX COEFFICIENTS

Consider the quadratic equation

$$ax^2 + bx + c = 0 \quad \dots(i)$$

where a, b, c are complex numbers and $a \neq 0$.

Proceeding as in section 14.4, we obtain that the roots of equation (i) are given by

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

These roots are complex as a, b, c are complex numbers.

Since the order relation is not defined in case of complex numbers. Therefore, we cannot assign positive or negative sign to the discriminate $D = b^2 - 4ac$. However, equation (i) has complex roots which are equal, if $D = b^2 - 4ac = 0$ and unequal roots if $D = b^2 - 4ac \neq 0$.

REMARK In case of quadratic equations with real coefficients imaginary (complex) roots always occur in conjugate pairs. However, it is not true for quadratic equations with complex coefficients. For example, the equation $4x^2 - 4ix - 1 = 0$ has both roots equal to $\frac{1}{2}i$.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Solve the following quadratic equations by factorization method:

(i) $x^2 - 5ix - 6 = 0$

(ii) $x^2 + 4ix - 4 = 0$

SOLUTION (i) The given equation is

$$x^2 - 5ix - 6 = 0$$

$$\Rightarrow x^2 - 5ix + 6i^2 = 0$$

$$\Rightarrow x^2 - 3ix - 2ix + 6i^2 = 0$$

$$\Rightarrow x(x - 3i) - 2i(x - 3i) = 0$$

$$\Rightarrow (x - 3i)(x - 2i) = 0$$

$$\Rightarrow x - 3i = 0, x - 2i = 0$$

$$\Rightarrow x = 3i, x = 2i$$

Hence, the roots of the given equation are $3i$ and $2i$.

(ii) We have,

$$x^2 + 4ix - 4 = 0$$

$$\Rightarrow x^2 + 4ix + 4i^2 = 0$$

$$\Rightarrow (x + 2i)^2 = 0$$

$$\Rightarrow x + 2i = 0 \quad (\text{twice})$$

$$\Rightarrow x = -2i, -2i$$

Hence, both the roots of the equation are equal to $-2i$.

EXAMPLE 2 Solve the following equations by factorization method

(i) $x^2 - \sqrt{2}ix + 12 = 0$ (ii) $3x^2 + 7ix + 6 = 0$ (iii) $x^2 - (3\sqrt{2} + 2i)x + 6\sqrt{2}i = 0$

SOLUTION (i) We have,

$$\begin{aligned} & x^2 - \sqrt{2}ix + 12 = 0 \\ \Rightarrow & x^2 - 3\sqrt{2}ix + 2\sqrt{2}ix - 12i^2 = 0 \\ \Rightarrow & x(x - 3\sqrt{2}i) + 2\sqrt{2}i(x - 3\sqrt{2}i) = 0 \\ \Rightarrow & (x - 3\sqrt{2}i)(x + 2\sqrt{2}i) = 0 \\ \Rightarrow & x - 3\sqrt{2}i = 0 \text{ or, } x + 2\sqrt{2}i = 0 \\ \Rightarrow & x = 3\sqrt{2}i \text{ or, } x = -2\sqrt{2}i \end{aligned}$$

Hence, the roots of the given equation are $-2\sqrt{2}i$ and $3\sqrt{2}i$.

(ii) $3x^2 + 7ix + 6 = 0$

$$\begin{aligned} \Rightarrow & 3x^2 + 9ix - 2ix - 6i^2 = 0 \\ \Rightarrow & 3x(x + 3i) - 2i(x + 3i) = 0 \\ \Rightarrow & (x + 3i)(3x - 2i) = 0 \\ \Rightarrow & x + 3i = 0 \text{ or, } 3x - 2i = 0 \\ \Rightarrow & x = -3i \text{ or, } x = \frac{2}{3}i \end{aligned}$$

Hence, the roots of the given equation are $-3i$ and $\frac{2}{3}i$.

(iii) $x^2 - (3\sqrt{2} + 2i)x + 6\sqrt{2}i = 0$

$$\begin{aligned} \Rightarrow & (x^2 - 3\sqrt{2}x) - (2ix - 6\sqrt{2}i) = 0 \\ \Rightarrow & x(x - 3\sqrt{2}) - 2i(x - 3\sqrt{2}) = 0 \\ \Rightarrow & (x - 2i)(x - 3\sqrt{2}) = 0 \\ \Rightarrow & x - 2i = 0, x - 3\sqrt{2} = 0 \\ \Rightarrow & x = 2i \text{ or, } 3\sqrt{2} \end{aligned}$$

Hence, the roots of the given equation are $2i$ and $3\sqrt{2}$.

EXAMPLE 3 Solve the following quadratic equations by using the general expressions for the roots of a quadratic equation:

(i) $x^2 - (3\sqrt{2} - 2i)x - 6\sqrt{2}i = 0$ (ii) $2x^2 + 3ix + 2 = 0$

SOLUTION (i) On comparing the given equation with the general equation $ax^2 + bx + c = 0$, we get: $a = 1$, $b = -(3\sqrt{2} - 2i)$ and $c = -6\sqrt{2}i$.

Substituting the values of a, b, c in $\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$, we get

$$\begin{aligned} \alpha &= \frac{(3\sqrt{2} - 2i) + \sqrt{(3\sqrt{2} - 2i)^2 + 24\sqrt{2}i}}{2}, \text{ and } \beta = \frac{(3\sqrt{2} - 2i) - \sqrt{(3\sqrt{2} - 2i)^2 + 24\sqrt{2}i}}{2} \\ \Rightarrow \alpha &= \frac{(3\sqrt{2} - 2i) + \sqrt{(3\sqrt{2} + 2i)^2}}{2}, \text{ and } \beta = \frac{(3\sqrt{2} - 2i) - \sqrt{13\sqrt{2} + 2i)^2}}{2} \\ \Rightarrow \alpha &= \frac{3\sqrt{2} - 2i + 3\sqrt{2} + 2i}{2}, \text{ and } \beta = \frac{(3\sqrt{2} - 2i) - (3\sqrt{2} + 2i)}{2} \\ \Rightarrow \alpha &= 3\sqrt{2}, \beta = -2i \end{aligned}$$

Hence, the roots of the given equation are $3\sqrt{2}$ and $-2i$.

(ii) On comparing the given equation with the general equation $ax^2 + bx + c = 0$, we get $a = 2, b = 3i$ and $c = 2$.

Substituting these values of a, b, c in $\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$, we get

$$\alpha = \frac{-3i + \sqrt{9i^2 - 16}}{4} \quad \text{and} \quad \beta = \frac{-3i - \sqrt{9i^2 - 16}}{4}$$

$$\Rightarrow \alpha = \frac{-3i + \sqrt{-25}}{4} \quad \text{and} \quad \beta = \frac{-3i - \sqrt{-25}}{4}$$

$$\Rightarrow \alpha = \frac{-3i + 5i}{4} \quad \text{and} \quad \beta = \frac{-3i - 5i}{4}$$

$$\Rightarrow \alpha = \frac{i}{2} \quad \text{and} \quad \beta = -2i$$

Hence, the roots of the given equation are $\frac{i}{2}$ and $-2i$.

LEVEL-2

EXAMPLE 4 Solve: $2x^2 - (3 + 7i)x - (3 - 9i) = 0$.

SOLUTION On comparing the given equation with the general form $ax^2 + bx + c = 0$, we obtain $a = 2, b = -(3 + 7i), c = -(3 - 9i)$.

Substituting the values of a, b, c in $\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$, we get

$$\alpha = \frac{(3 + 7i) + \sqrt{(3 + 7i)^2 + 8(3 - 9i)}}{4} \quad \text{and} \quad \beta = \frac{(3 + 7i) - \sqrt{(3 + 7i)^2 + 8(3 - 9i)}}{4}$$

$$\Rightarrow \alpha = \frac{(3 + 7i) + \sqrt{9 - 49 + 42i + 24 - 72i}}{4} \quad \text{and} \quad \beta = \frac{(3 + 7i) - \sqrt{9 - 49 + 42i + 24 - 72i}}{4}$$

$$\Rightarrow \alpha = \frac{3 + 7i + \sqrt{-16 - 30i}}{4} \quad \text{and} \quad \beta = \frac{3 + 7i - \sqrt{-16 - 30i}}{4} \quad \dots(i)$$

Let us now find $\sqrt{-16 - 30i}$.

Let $a + ib = \sqrt{-16 - 30i}$. Then,

$$a + ib = \sqrt{-16 - 30i}$$

$$\Rightarrow a^2 - b^2 + 2iab = -16 - 30i$$

$$\Rightarrow a^2 - b^2 = -16 \quad \dots(ii)$$

$$\text{and, } 2ab = -30 \quad \dots(iii)$$

$$\therefore (a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$$

$$\Rightarrow (a^2 + b^2)^2 = 256 + 900 = 1156$$

$$\Rightarrow a^2 + b^2 = 34$$

$$\text{Now, } a^2 - b^2 = -16 \text{ and } a^2 + b^2 = 34$$

$$\Rightarrow a^2 = 9 \text{ and } b^2 = 25$$

$$\Rightarrow a = \pm 3 \text{ and } b = \pm 5$$

From (iii), we find that a and b are of opposite signs.

$\therefore a = 3 \text{ and } b = -5 \text{ or, } a = -3 \text{ and } b = 5.$

Hence, $\sqrt{-16 - 30i} = 3 - 5i$ or, $-3 + 5i$.

Substituting either of these values in (i), we get

$$\alpha = \frac{(3 + 7i) + (3 - 5i)}{4} \text{ and, } \beta = \frac{(3 + 7i) - (3 - 5i)}{4}$$

$$\Rightarrow \alpha = \frac{3}{2} + \frac{1}{2}i \text{ and, } \beta = 3i$$

Hence, the roots of the given equation are $\frac{3}{2} + \frac{1}{2}i$ and $3i$

EXAMPLE 5 Solve: $x^2 - (7 - i)x + (18 - i) = 0$ over C .

SOLUTION Comparing the given equation with the general form $ax^2 + bx + c = 0$, we get $a = 1$, $b = -(7 - i)$ and $c = 18 - i$. Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and, } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{(7 - i) + \sqrt{(7 - i)^2 - 4(18 - i)}}{2}, \quad \beta = \frac{(7 - i) - \sqrt{(7 - i)^2 - 4(18 - i)}}{2}$$

$$\Rightarrow \alpha = \frac{(7 - i) + \sqrt{-24 - 10i}}{2}, \quad \beta = \frac{(7 - i) - \sqrt{-24 - 10i}}{2} \quad \dots(i)$$

Let us now find $\sqrt{-24 - 10i}$.

Let, $a + ib = \sqrt{-24 - 10i}$. Then,

$$(a + ib)^2 = -24 - 10i$$

$$\Rightarrow (a^2 - b^2) + 2i ab = -24 - 10i$$

$$\Rightarrow a^2 - b^2 = -24 \quad \dots(ii)$$

$$\text{and, } 2ab = -10 \quad \dots(iii)$$

$$\text{Now, } (a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$$

$$\Rightarrow (a^2 + b^2)^2 = 576 + 100 = 676 \Rightarrow a^2 + b^2 = 26 \quad \dots(iv)$$

Solving (ii) and (iii), we get $a = \pm 1$ and $b = \pm 5$.

From (iii), we find that ab is negative.

$$\therefore a = 1, b = -5 \text{ or, } a = -1, b = 5.$$

$$\therefore a + ib = 1 - 5i \text{ or, } -1 + 5i$$

$$\text{Hence, } \sqrt{-24 - 10i} = \pm(1 - 5i)$$

Substituting either of these values in (i), we get

$$\alpha = \frac{7 - i + 1 - 5i}{2} = 4 - 3i \text{ and } \beta = \frac{(7 - i) - (1 - 5i)}{2} = 3 + 2i$$

Hence, the roots of the given equation are $4 - 3i$ and $3 + 2i$.

EXERCISE 14.2

1. Solving the following quadratic equations by factorization method:

(i) $x^2 + 10ix - 21 = 0$

(ii) $x^2 + (1 - 2i)x - 2i = 0$

(iii) $x^2 - (2\sqrt{3} + 3i)x + 6\sqrt{3}i = 0$

(iv) $6x^2 - 17ix - 12 = 0$

2. Solve the following quadratic equations:

(i) $x^2 - (3\sqrt{2} + 2i)x + 6\sqrt{2}i = 0$

(ii) $x^2 - (5 - i)x + (18 + i) = 0$

(iii) $(2 + i)x^2 - (5 - i)x + 2(1 - i) = 0$

(iv) $x^2 - (2 + i)x - (1 - 7i) = 0$

(v) $ix^2 - 4x - 4i = 0$

(vi) $x^2 + 4ix - 4 = 0$

- (vii) $2x^2 + \sqrt{15}ix - i = 0$ [NCERT] (viii) $x^2 - x + (1+i) = 0$
 (ix) $ix^2 - x + 12i = 0$ [NCERT] (x) $x^2 - (3\sqrt{2} - 2i)x - \sqrt{2}i = 0$
 (xi) $x^2 - (\sqrt{2} + i)x + \sqrt{2}i = 0$ [NCERT] (xii) $2x^2 - (3+7i)x + (9i-3) = 0$

ANSWERS

1. (i) $-3i, -7i$ (ii) $-1, 2i$ (iii) $2\sqrt{3}, 3i$ (iv) $\frac{3}{2}i, \frac{4}{3}i$
 2. (i) $3\sqrt{2}, 2i$ (ii) $3-4i, 2+3i$ (iii) $1-i, \frac{4}{5}-\frac{2}{5}i$ (iv) $3-i, -1+2i$
 (v) $-2i, -2i$ (vi) $-2i, -2i$ (vii) $\frac{1+(4-\sqrt{15})i}{4}, \frac{-1-(\sqrt{15}+4)i}{4}$
 (viii) $1-i, i$ (ix) $-4i, 3i$ (x) $\frac{3\sqrt{2}-2i}{2} \pm \frac{4-\sqrt{2}i}{2}$
 (xi) $\sqrt{2}, i$ (xii) $\frac{3+i}{2}, 3i$

HINTS TO NCERT & SELECTED PROBLEMS

2. (vii) The given equation is $2x^2 + \sqrt{15}ix - i = 0$.

Comparing this equation with the standard equation $ax^2 + bx + c = 0$, we get

$$a = 2, b = \sqrt{15}i \text{ and } c = -i$$

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{-\sqrt{15}i + \sqrt{-15+8i}}{4} \text{ and } \beta = \frac{-\sqrt{15}i - \sqrt{-15+8i}}{4}$$

Let $\sqrt{-15+8i} = a+ib$. Then,

$$-15+8i = (a+ib)^2$$

$$\Rightarrow -15+8i = a^2 - b^2 + 2iab$$

$$\Rightarrow a^2 - b^2 = -15 \text{ and } 2ab = 8$$

$$\text{Now, } (a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$$

$$\Rightarrow (a^2 + b^2)^2 = (-15)^2 + 64 = 289$$

$$\Rightarrow a^2 + b^2 = 17$$

Solving $a^2 - b^2 = -15$ and $a^2 + b^2 = 17$, we get

$$a^2 = 1 \text{ and } b^2 = 16$$

$$\Rightarrow a = \pm 1 \text{ and } b = \pm 4$$

$$\Rightarrow a = 1, b = 4 \text{ or } a = -1, b = -4 \quad [\because ab = 4 > 0 \therefore a \text{ and } b \text{ are of the same sign}]$$

$$\therefore \sqrt{-15+8i} = 1+4i, -1-4i$$

$$\text{When } \sqrt{-15+8i} = 1+4i:$$

$$\alpha = \frac{\sqrt{15}i + 1 + 4i}{4} = \frac{1 + (\sqrt{15} + 4)i}{4} \text{ and } \beta = \frac{-\sqrt{15}i - (1 + 4i)}{2} = -\frac{1 - (\sqrt{5} + 4)i}{2}$$

When $\sqrt{-15 + 8i} = -1 - 4i$:

$$\alpha = \frac{\sqrt{15}i - 1 - 4i}{2} = \frac{-1 - (\sqrt{15} + 4)i}{2} \text{ and } \beta = \frac{-\sqrt{15}i + 1 + 4i}{2} = \frac{1 + (4 - \sqrt{15})i}{2}$$

(ix) The given equation is $ix^2 - x + 12i = 0$.

Comparing this equation with the standard equation $ax^2 + bx + c = 0$, we get $a = i, b = -1$ and $c = 12i$.

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{1 + \sqrt{1 + 48}}{2i} \text{ and } \beta = \frac{1 - \sqrt{1 + 48}}{2i}$$

$$\Rightarrow \alpha = \frac{1 + 7}{2i} \text{ and } \beta = \frac{1 - 7}{2i}$$

$$\Rightarrow \alpha = \frac{4}{i} \text{ and } \beta = -\frac{3}{i} \Rightarrow \alpha = 0 - 4i \text{ and } \beta = 3i$$

(xi) The given equation is $x^2 - (\sqrt{2} + i)x + \sqrt{2}i = 0$.

Comparing this equation with the standard equation $ax^2 + bx + c = 0$, we get $a = 1, b = -(\sqrt{2} + i)$ and $c = \sqrt{2}i$.

Substituting these values in

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}, \text{ we get}$$

$$\alpha = \frac{(\sqrt{2} + i) + \sqrt{(\sqrt{2} + i)^2 - 4\sqrt{2}i}}{2} \text{ and } \beta = \frac{(\sqrt{2} + i) - \sqrt{(\sqrt{2} + i)^2 - 4\sqrt{2}i}}{2}$$

$$\Rightarrow \alpha = \frac{\sqrt{2} + i + \sqrt{(\sqrt{2} - i)^2}}{2} \text{ and } \beta = \frac{(\sqrt{2} + i) - \sqrt{(\sqrt{2} - i)^2}}{2}$$

$$\Rightarrow \alpha = \frac{\sqrt{2} + i + \sqrt{2} - i}{2} \text{ and } \beta = \frac{(\sqrt{2} + i) - (\sqrt{2} - i)}{2}$$

$$\Rightarrow \alpha = \sqrt{2} \text{ and } \beta = i$$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

1. Write the number of real roots of the equation $(x - 1)^2 + (x - 2)^2 + (x - 3)^2 = 0$.
2. If a and b are roots of the equation $x^2 - px + q = 0$, then write the value of $\frac{1}{a} + \frac{1}{b}$.
3. If roots α, β of the equation $x^2 - px + 16 = 0$ satisfy the relation $\alpha^2 + \beta^2 = 9$, then write the value of p .
4. If $2 + \sqrt{3}$ is a root of the equation $x^2 + px + q = 0$, then write the values of p and q .

5. If the difference between the roots of the equation $x^2 + ax + 8 = 0$ is 2, write the values of a .
6. Write the roots of the equation $(a-b)x^2 + (b-c)x + (c-a) = 0$.
7. If a and b are roots of the equation $x^2 - x + 1 = 0$, then write the value of $a^2 + b^2$.
8. Write the number of quadratic equations, with real roots, which do not change by squaring their roots.
9. If α, β are roots of the equation $x^2 + lx + m = 0$, write an equation whose roots are $-\frac{1}{\alpha}$ and $-\frac{1}{\beta}$.
10. If α, β are roots of the equation $x^2 - a(x+1) - c = 0$, then write the value of $(1+\alpha)(1+\beta)$.

ANSWERS

1. No real root 2. $\frac{1}{q}$ 3. ± 8 4. $p = -4, q = 1$ 5. ± 8 6. $1, \frac{c-a}{a-b}$
 7. -1 8. 3 9. $mx^2 - lx + 1 = 0$ 10. $1 - c$

MULTIPLE CHOICE QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

1. The complete set of values of k , for which the quadratic equation $x^2 - kx + k + 2 = 0$ has equal roots, consists of
 (a) $2 + \sqrt{12}$ (b) $2 \pm \sqrt{12}$ (c) $2 - \sqrt{12}$ (d) $-2 - \sqrt{2}$
2. For the equation $|x|^2 + |x| - 6 = 0$, the sum of the real roots is
 (a) 1 (b) 0 (c) 2 (d) none of these
3. If a, b are the roots of the equation $x^2 + x + 1 = 0$, then $a^2 + b^2 =$
 (a) 1 (b) 2 (c) -1 (d) 3
4. If α, β are roots of the equation $4x^2 + 3x + 7 = 0$, then $1/\alpha + 1/\beta$ is equal to
 (a) $7/3$ (b) $-7/3$ (c) $3/7$ (d) $-3/7$
5. The values of x satisfying $\log_3(x^2 + 4x + 12) = 2$ are
 (a) $2, -4$ (b) $1, -3$ (c) $-1, 3$ (d) $-1, -3$
6. The number of real roots of the equation $(x^2 + 2x)^2 - (x+1)^2 - 55 = 0$ is
 (a) 2 (b) 1 (c) 4 (d) none of these
7. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b} =$
 (a) c/ab (b) a/bc (c) b/ac (d) none of these
8. If α, β are the roots of the equation $x^2 + px + 1 = 0$; γ, δ the roots of the equation $x^2 + qx + 1 = 0$, then $(\alpha - \gamma)(\alpha + \delta)(\beta - \gamma)(\beta + \delta) =$
 (a) $q^2 - p^2$ (b) $p^2 - q^2$ (c) $p^2 + q^2$ (d) none of these
9. The number of real solutions of $|2x - x^2 - 3| = 1$ is
 (a) 0 (b) 2 (c) 3 (d) 4

10. The number of solutions of $x^2 + |x - 1| = 1$ is
 (a) 0 (b) 1 (c) 2 (d) 3
11. If x is real and $k = \frac{x^2 - x + 1}{x^2 + x + 1}$, then
 (a) $k \in [1/3, 3]$ (b) $k \geq 3$ (c) $k \leq 1/3$ (d) none of these
12. If the roots of $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c$ is
 (a) 0 (b) 1 (c) 2 (d) none of these
13. The value of a such that $x^2 - 11x + a = 0$ and $x^2 - 14x + 2a = 0$ may have a common root is
 (a) 0 (b) 12 (c) 24 (d) 32.
14. The values of k for which the quadratic equation $kx^2 + 1 = kx + 3x - 11x^2$ has real and equal roots are
 (a) $-11, -3$, (b) $5, 7$ (c) $5, -7$ (d) none of these
15. If the equations $x^2 + 2x + 3\lambda = 0$ and $2x^2 + 3x + 5\lambda = 0$ have a non-zero common roots, then $\lambda =$
 (a) 1 (b) -1 (c) 3 (d) none of these
16. If one root of the equation $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, the value of q is
 (a) $49/4$ (b) $4/49$ (c) 4 (d) none of these
17. The value of p and q ($p \neq 0, q \neq 0$) for which p, q are the roots of the equation $x^2 + px + qab = 0$ are
 (a) $p = 1, q = -2$ (b) $p = -1, q = -2$
 (c) $p = -1, q = 2$ (d) $p = 1, q = 2$
18. The set of all values of m for which both the roots of the equation $x^2 - (m + 1)x + m + 4 = 0$ are real and negative, is
 (a) $(-\infty, -3] \cup [5, \infty)$ (b) $[-3, 5]$
 (c) $(-4, -3]$ (d) $(-3, -1]$
19. The number of roots of the equation $\frac{(x+2)(x-5)}{(x-3)(x+6)} = \frac{x-2}{x+4}$ is
 (a) 0 (b) 1 (c) 2 (d) 3
20. If α and β are the roots of $4x^2 + 3x + 7 = 0$, then the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is
 (a) $\frac{4}{7}$ (b) $-\frac{3}{7}$ (c) $\frac{3}{7}$ (d) $-\frac{3}{4}$
21. If α, β are the roots of the equation $x^2 + px + q = 0$, then $-\frac{1}{\alpha}, -\frac{1}{\beta}$ are the roots of the equation
 (a) $x^2 - px + q = 0$ (b) $x^2 + px + q = 0$
 (c) $qx^2 + px + 1 = 0$ (d) $qx^2 - px + 1 = 0$
22. If the difference of the roots of $x^2 - px + q = 0$ is unity, then
 (a) $p^2 + 4q = 1$ (b) $p^2 - 4q = 1$
 (c) $p^2 + 4q^2 = (1 + 2q)^2$ (d) $4p^2 + q^2 = (1 + 2p)^2$

23. If α, β are the roots of the equation $x^2 - p(x+1) - c = 0$, then $(\alpha + 1)(\beta + 1) =$
 (a) c (b) $c - 1$ (c) $1 - c$ (d) none of these
24. The least value of k which makes the roots of the equation $x^2 + 5x + k = 0$ imaginary is
 (a) 4 (b) 5 (c) 6 (d) 7
25. The equation of the smallest degree with real coefficients having $1 + i$ as one of the roots is
 (a) $x^2 + x + 1 = 0$ (b) $x^2 - 2x + 2 = 0$
 (c) $x^2 + 2x + 2 = 0$ (d) $x^2 + 2x - 2 = 0$

ANSWERS

1. (b) 2. (b) 3. (c) 4. (d) 5. (d) 6. (b) 7. (c) 8. (a)
 9. (b) 10. (a) 11. (a) 12. (b) 13. (c) 14. (c) 15. (b) 16. (a)
 17. (c) 18. (a) 19. (b) 20. (b) 21. (d) 22. (b), (c) 23. (c)
 24. (d) 25. (b)

SUMMARY

1. Fundamental Theorem of Algebra: Every polynomial equation $f(x) = 0$ has at least one root, real or imaginary (complex).
2. Every polynomial equation $f(x) = 0$ of degree n has exactly n roots real or imaginary.
3. A quadratic equation cannot have more than two roots.
4. If $ax^2 + bx + c = 0, a \neq 0$ is a quadratic equation with real coefficients, then its roots α and β given by

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \text{ or } \alpha = \frac{-b + \sqrt{D}}{2a} \text{ and } \beta = \frac{-b - \sqrt{D}}{2a}$$

where $D = b^2 - 4ac$ is as the discriminant of the equation.

(i) If $D = 0$, then $\alpha = \beta = -\frac{b}{2a}$

So, the equation has real and equal roots each equal to $-\frac{b}{2a}$.

- (ii) If $a, b, c \in \mathbb{Q}$ and D is positive and a perfect square, then roots are rational and unequal.
- (iii) If $a, b, c \in \mathbb{R}$ and D is positive and a perfect square, then the roots are real and distinct.
- (iv) If $D > 0$ but it is not a perfect square, then roots are irrational and unequal.
- (v) If $D < 0$, then the roots are imaginary and are given by

$$\alpha = \frac{-b + i\sqrt{4ac - b^2}}{2a} \text{ and } \beta = \frac{-b - i\sqrt{4ac - b^2}}{2a}$$

- (vi) If $a = 1, b, c \in \mathbb{I}$ and the roots are rational numbers, then these roots must be integers.
- (vii) If a quadratic equation in x has more than two roots, then it is an identity in x that is $a = b = c = 0$.
- (viii) Complex roots of an equation with real coefficients always occur in pairs. However, this may not be true in case of equations with complex coefficients. For example, $x^2 - 2ix - 1 = 0$ has both roots equal to i .
- (ix) Surd root of an equation with rational coefficients always occur in pairs like $2 + \sqrt{3}$ and $2 - \sqrt{3}$. However, Surd roots of an equation with irrational coefficients may not occur in pairs. For example, $x^2 - 2\sqrt{3}x + 3 = 0$ has both roots equal to $\sqrt{3}$.

LINEAR INEQUATIONS

15.1 INTRODUCTION

In this chapter, we will study linear inequations in one and two variables. The knowledge of linear inequations is very helpful in solving problems in Science, Mathematics, Engineering, Linear Programming etc.

15.2 INEQUATIONS

In earlier classes, we have studied equations in one and two variables. An equation is defined as a statement involving variable (s) and the sign of equality ($=$). Similarly, we define the term inequation as follows:

INEQUATION A statement involving variable (s) and the sign of inequality viz, $>$, $<$, $\geq$ or $\leq$ is called an inequation or an inequality.

An inequation may contain one or more variables. Also, it may be linear or quadratic or cubic etc.

Following are some examples of inequations:

- | | | |
|-----------------------------------|-------------------------------------|-----------------------------|
| (i) $3x - 2 < 0$ | (ii) $2x + 3 \leq 0$ | (iii) $5x - 3 > 0$ |
| (iv) $4x + 5 \geq 0$ | (v) $2x + 3y < 1$ | (vi) $5x + 4y \leq 3$ |
| (vii) $4x - 6y > 5$ | (viii) $2x + 5y \geq 4$ | (ix) $2x^2 + 3x + 4 > 0$ |
| (x) $x^2 - 3x + 2 \geq 0$ | (xi) $x^2 + 3x + 2 < 0$ | (xii) $x^2 - 5x + 4 \leq 0$ |
| (xiii) $x^3 - 6x^2 + 11x - 6 > 0$ | (xiv) $x^3 + 6x^2 + 11x + 6 \leq 0$ | |

LINEAR INEQUATION IN ONE VARIABLE Let a be a non-zero real number and x be a variable. Then inequations of the form $ax + b < 0$, $ax + b \leq 0$, $ax + b > 0$ and $ax + b \geq 0$ are known as linear inequations in one variable x .

For example, $9x - 15 > 0$, $5x - 4 \geq 0$, $3x + 2 < 0$ and $2x - 3 \leq 0$ are linear inequations in one variable.

LINEAR INEQUATIONS IN TWO VARIABLES Let a, b be non-zero real numbers and x, y be variables. Then inequations of the form $ax + by < c$, $ax + by \leq c$, $ax + by > c$ and $ax + by \geq c$ are known as linear inequations in two variables x and y .

For example, $2x + 3y \leq 6$, $3x - 2y \geq 12$, $x + y < 4$, $2x + y \geq 6$ are linear inequations in two variables x and y .

QUADRATIC INEQUATION Let a be a non-zero real number. Then an inequation of the form $ax^2 + bx + c < 0$, or $ax^2 + bx + c \leq 0$, or $ax^2 + bx + c > 0$, or $ax^2 + bx + c \geq 0$ is known as a quadratic inequation.

For example, $x^2 + x - 6 < 0$, $x^2 - 3x + 2 \geq 0$, $2x^2 + 3x + 1 > 0$ and $x^2 - 5x + 4 \leq 0$ are quadratic inequations.

In this chapter, we shall study linear inequations in one and two variables only.

15.3 SOLUTIONS OF AN INEQUATION

DEFINITION A solution of an inequation is the value (s) of the variable (s) that makes it a true statement.

Consider the inequation $\frac{3-2x}{5} < \frac{x}{3} - 4$.

Left hand side (LHS) of this inequation is $\frac{3-2x}{5}$ and right hand side (RHS) is $\frac{x}{3} - 4$.

We observe that:

For $x = 9$, we have

$$\text{LHS} = \frac{3-2 \times 9}{5} = -3 \text{ and, RHS} = \frac{9}{3} - 4 = -1$$

Clearly, $-3 < -1$

$\Rightarrow$ LHS < RHS, which is true.

So, $x = 9$ is a solution of the given inequation.

For $x = 6$, we have

$$\text{LHS} = \frac{3-2 \times 6}{5} = -\frac{9}{5} \text{ and RHS} = \frac{6}{3} - 4 = -2$$

Because, $-\frac{9}{5} < -2$ is not true. So, $x = 6$ is not a solution of the given inequation.

We can verify that any real number greater than 7 is a solution of the given inequation.

Let us now consider the inequation $x^2 + 1 < 0$.

We know that

$$x^2 \geq 0 \text{ for all } x \in R$$

$$\therefore x^2 + 1 \geq 1 \text{ for all } x \in R$$

$$\Rightarrow x^2 + 1 \nless 0 \text{ for any } x \in R.$$

So, there is no real value of x which makes the given inequation a true statement. Hence, it has no solution.

It follows from the above discussion that an inequation may or may not have a solution. However, if an inequation has a solution it may have infinitely many solutions.

SOLVING AN INEQUATION It is the process of obtaining all possible solutions of an inequation.

SOLUTION SET The set of all possible solutions of an inequation is known as its solution set.

For example, the solution set of the inequation $x^2 + 1 \geq 0$ is the set R of all real numbers whereas the solution set of the inequation $x^2 + 1 < 0$ is the null set ϕ .

15.4 SOLVING LINEAR INEQUATIONS IN ONE VARIABLE

As mentioned in the previous section that solving an inequation is the process of obtaining its all possible solutions. In the process of solving an inequation, we use mathematical simplifications which are governed by the following rules:

RULE 1 Same number may be added to (or subtracted from) both sides of an inequation without changing the sign of inequality.

RULE 2 Both sides of an inequation can be multiplied (or divided) by the same positive real number without changing the sign of inequality. However, the sign of inequality is reversed when both sides of an inequation are multiplied or divided by a negative number.

RULE 3 Any term of an inequation may be taken to the other side with its sign changed without affecting the sign of inequality.

A linear inequation in one variable is of the form

$$ax + b < 0 \text{ or, } ax + b \leq 0 \text{ or, } ax + b > 0 \text{ or, } ax + b \geq 0.$$

We follow the following algorithm to solve a linear inequation in one variable.

ALGORITHM

STEP I Obtain the linear inequation.

STEP II Collect all terms involving the variable on one side of the inequation and the constant terms on the other side.

STEP III Simplify both sides of inequality in their simplest forms to reduce the inequation in the form $ax < b$, or $ax \leq b$, or $ax > b$, or $ax \geq b$

STEP IV Solve the inequation obtained in step III by dividing both sides of the inequation by the coefficient of the variable.

STEP V Write the solution set obtained in step IV in the form of an interval on the real line.

Following examples will illustrate the above algorithm.

ILLUSTRATIVE EXAMPLES

LEVEL-1

Type I SOLVING EQUATIONS OF THE FORM: $ax + b > cx + d$, or, $ax + b \geq cx + d$,
or, $ax + b < cx + d$ or, $ax + b \leq cx + d$

EXAMPLE 1 Solve the following linear inequations:

(i) $2x - 4 \leq 0$ (ii) $-3x + 12 < 0$ (iii) $4x - 12 \geq 0$ (iv) $7x + 9 > 30$

SOLUTION (i) We have,

$$2x - 4 \leq 0$$

$$\Rightarrow (2x - 4) + 4 \leq 0 + 4$$

[Adding 4 on both sides]

$$\Rightarrow 2x \leq 4$$

$$\Rightarrow \frac{2x}{2} \leq \frac{4}{2}$$

$$\Rightarrow x \leq 2$$

Hence, any real number less than or equal to 2 is a solution of the given inequation.

These solutions can be graphed on real line as shown in Fig. 15.1

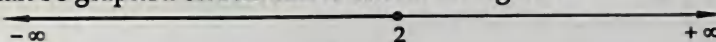


Fig. 15.1

The solution set of the given inequation is $(-\infty, 2]$

(ii) We have,

$$-3x + 12 < 0$$

$$\Rightarrow -3x < -12$$

[Transposing 12 on right side]

$$\Rightarrow \frac{-3x}{-3} > \frac{-12}{-3}$$

[Dividing both sides by -3]

$$\Rightarrow x > 4$$

Thus, any real number greater than 4 is a solution of the given inequation.

Hence, the solution set of the given inequation is $(4, \infty)$. This solution set can be graphed on real line as shown in Fig. 15.2

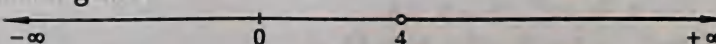


Fig. 15.2

(iii) We have,

$$4x - 12 \geq 0$$

$$\begin{aligned}
 \Rightarrow 4x &\geq 12 && \text{[Transposing 12 on RHS]} \\
 \Rightarrow \frac{4x}{4} &\geq \frac{12}{4} && \text{[Dividing both sides by 4]} \\
 \Rightarrow x &\geq 3 \\
 \Rightarrow x &\in [3, \infty)
 \end{aligned}$$

Hence, the solution set of the given inequation is $[3, \infty)$. This solution set can be graphed on real line as shown in Fig. 15.3



Fig. 15.3

(iv) We have,

$$\begin{aligned}
 7x + 9 &> 30 \\
 \Rightarrow 7x &> 30 - 9 \\
 \Rightarrow 7x &> 21 && \text{[Transposing 9 on RHS]} \\
 \Rightarrow \frac{7x}{7} &> \frac{21}{7} \\
 \Rightarrow x &> 3 \\
 \Rightarrow x &\in (3, \infty)
 \end{aligned}$$

Hence, $(3, \infty)$ is the solution set of the given inequation. This can be graphed on real line as shown in Fig. 15.4.



Fig. 15.4

EXAMPLE 2 Solve: $5x - 3 < 3x + 1$ when (i) x is a real number (ii) x is integer number (iii) x is a natural number.

SOLUTION We have,

$$\begin{aligned}
 5x - 3 &< 3x + 1 \\
 \Rightarrow 5x - 3x &< 3 + 1 && \text{[Transposing } 3x \text{ on LHS and } -3 \text{ on RHS]} \\
 \Rightarrow 2x &< 4 \\
 \Rightarrow \frac{2x}{2} &< \frac{4}{2} && \left[\text{Multiplying both sides by } \frac{1}{2} \right] \\
 \Rightarrow x &< 2
 \end{aligned}$$

(i) If $x \in \mathbb{R}$, then

$$x < 2 \Rightarrow x \in (-\infty, 2)$$

Hence, the solution set is $(-\infty, 2)$ as shown in Fig. 15.5.

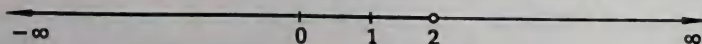


Fig. 15.5

(ii) If $x \in \mathbb{Z}$, then

$$x < 2 \Rightarrow x = 1, 0, -1, -2, -3, -4, \dots$$

So, the solution set is $\{\dots, -4, -3, -2, -1, 0, 1\}$

(iii) If $x \in \mathbb{N}$, then

$$x < 2 \Rightarrow x = 1$$

So, the solution set is $\{1\}$.